Forecasting the impact of systematic uncertainties on the dispersion mseasure redshift relation of fast radio bursts

Elyas Farah

Masterarbeit in Astrophysik
angefertigt im Argelander-Institut für Astronomie

vorgelegt der
Mathematisch-Naturwissenschaftlichen Fakultät
der
Rheinischen Friedrich-Wilhelms-Universität
Bonn

May 2026

I hereby declare that this thesis was formulated by myself and that no sources or tools other than those cited were used.

Bonn, ...

Date Signature

Gutachterin und Betreuerin: Prof. Dr. Jennifer Schöber
Gutachter: Prof. Dr. Cristriano Poriciani

Contents

1 Introduction **2**

2 Theoretical Background **6**
2.1 The general theory of relativity . . . 6
2.2 The Friedmann-Robertson-Lemaître-Walker metric . . . 9
2.3 The cosmological model . . . 12
2.4 The Boltzmann equation & the linear perturbation theory . . . 16
2.5 The matter power spectrum and dark matter tracers . . . 19
2.6 The halo model & the electron power spectrum . . . 20
2.7 Weak gravitational Lensing & cosmic shear . . . 22
2.8 The impact of baryonic feedback on the large-scale structure . . . 22
2.9 Fast radio bursts . . . 22

3 Methodology **25**
3.1 Modelling the electron power spectrum . . . 25
3.1.1 The baryonification correction model . . . 25
3.1.2 The halo model . . . 25
3.2 Simulating the dispersion measure signal . . . 25
3.2.1 The Generator for Large Scale Structure (GLASS) . . . 25
3.2.2 The electron angular power spectrum . . . 26
3.2.3 The cosmological covariance of the dispersion measure signal . . . 26
3.3 The systematic uncertainties . . . 26
3.3.1 Redshift uncertainty . . . 26
3.3.2 Selection effects . . . 26
3.3.3 Unlocalised hosts . . . 26
3.3.4 Milkyway contribution . . . 26
3.3.5 Host galaxy contribution . . . 26
3.4 Simulation-based inference . . . 26
3.4.1 Approximate Baysian inference . . . 26
3.4.2 Neural Posterior Estimator . . . 26
3.4.3 The Sequentual Neural Posterio Estimator . . . 26
3.4.4 Flow matching posterior estimator . . . 26
3.4.5 Data Compression . . . 26

4 Results **27**

5 Discussion & Conclusion 28

A Useful information 29

Bibliography 30

List of Figures 34

List of Tables 35

This is your thesis abstract. It may be in a language that is different from the rest of your thesis.

This is your thesis abstract. It may be in a language that is different from the rest of your thesis.

CHAPTER 1

Introduction

The questions on how did the universe came to be, why is the universe as such, and what is the universe heading towards falls upon cosmology to answer. Cosmology is the study of the universe on large scales in an attempt to answer the latter questions cite maybe Peebles. This pursuit started as a data-starved research cite and transformed into a data-rich endevour as a result of a platitutde of imaging surveys such as the LSST Vera Rubin observatory, Euclid, and Nancy Roman cite. The qualitative questions, with which this paragraph started,< transcribe in the statistical sense to be what is the cosmological model that best describes the universe on throughout length scales and energy scales.

The first attempt to describe the universe as a whole came through the lens of the general theory of relativity where Robertson and Walker cite built on the work of Friedmann and Lemaître to describe geometry of the universe through what is now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. We discuss this metric further in 2.2. This metric was largely motivated by the observations of Edwin Hubble who saw that galaxy further away from the Earth are moving faster than galaxies closer to Earth Hubble, 1929. Figure shows the plot from Hubble's work in 1929.

Further observations of the magnitude of distant galaxies have shown that the universe is expanding cite. The general theory of relativity attributes this expansion to the energy density content of the universe. Therefore, the question of what is driving the expansion of the universe and what are the componenets that contribute to the energy budget of the universe represent a crucial question in cosmology for which multiple cosmological models have been suggested (Di Valentino et al., 2025). In fact, the cosmological model has been motivated by both cosmological and astrophysical observation. Some of these observation are the measurement of galaxy rotation curves by Vera Rubin cite that hinted at an additional matter component that does not interact with light and the discovery of the cosmic microwave background (CMB) radiation that hinted at hot and homogeneous early universe.cite. Furthermore,early observaiton of clusters of galaxies have shown that the spatial distribution of galaxy deviates from randomness. This was shown by Peebles cite. By calculating the two point correlation function, there was clearly a signal in the way that galaxies are distributed which points out to an underlying mechanism in the formation of these galaxies (Totsuji and Kihara, 1969). The development of obsrevational techniques and instrumentation allowed the launch of the Center for Astrophysics Redshift Survey (Davis, Huchra et al., 1982) which showed that galaxy positions are correlated on large scales and that there is a structure in the universe (Davis and Peebles, 1983).

In the early 1990s, cosmology experienced a paradigm shift with the launch of the Cosmic Background Explorer (COBE) sattelite (Boggess et al., 1992). The COBE mission provided the first detection of

primordial CMB anisotropies and further confirmed that the universe is flat (Smoot, 1999). In the context of the FLRW metric, the flatness of the universe requires a total sum of the relative density parameters to be 1 (within error bounds). Therefore, in addition to the matter component (dark and baryons), there should be a third component that can account for the missing energy density. Using supernova data from the Hubble Space Telescope (Bahcall and Spitzer, 1982), A. G. Riess, Filippenko et al. (1998) found a strong evidence for an expanding universe universe with negative acceleration and supported the findings of the COBE mission for the existance of additional density component different from matter which is driving the expansion of the universe. Hereafter, the term 'dark energy was coined to refer to this component (Turner, 2001). Wiht this, the ΛCDM model was born, and the efforts to obtain a precise estimates of the cosmological parameters intensified (Bennett et al., 2003; Pope, 2004).

While early experiements relied on CMB, galaxy number counts, or supernova to infer cosmological parameters, gravitational lensing was in development as a tool to constrain the cosmological parameters. Gravitational lensing is the distortions of light geodesics by gravity when light travels from a distant source travels through the inhomogeneous universe towards an observer on the Earth (P. Schneider, Ehlers and Falco, 1992). On the level, of a galaxy, gravitional lensing effects manifest themselves as a distortion of the shape of the galaxy and a change in the angular position with respect to a particular observer. Kaiser and Squires (1993) showed that these distortions could be used to obtain a map of the integrated mass density along the line of sight. The use of galaxy shape distortions for cosmology was a direct result of the Kairser-Squire inversion formula where Seitz and P. Schneider (1996) realised that the expectation value of galaxy shape is an unbiased estimate of the shear in the weak field limit. In principle, it is possible to relate the power spectrum of galaxy shapes with the matter power spectrum, offering a concrete method of probing cosmology by studying the correlations between galaxy shapes (Bartelmann and P. Schneider, 2001). This new probe of the large scale structure is called cosmic shear and was used by Maoli et al. (2001) to obtain constraints on the cosmological parameters. These constraints were not as tight as the ones obtained by COBE, however, it paved the road towards using cosmic shear to constrain the cosmological model. The advantage of cosmic shear as an integrated quantitity along the line of sight mean that it is sensitive to structure growth in the late universe. Therefore, it represent a very good tool to constrain the equation of state for dark energy (Hu and Tegmark, 1999; Jain and Taylor, 2003).

After that, multiple galaxy surveys used cosmic shear for cosmilogical parameter inference, and numerous statistical tools were developped to deal with the increasing data size (P. Schneider, Van Waerbeke et al., 1998; P. Schneider, Eifler and Krause, 2010). Likewise, with increasing data size, the noise level in a cosmological signal increases and the need to mititgate the impact of the systematic and measreument uncertainties is imperative. In this work, we focus on tackling on of the most crucial systematics, namely baryonic feedback on the large scale structure.

Baryonic feedback are the set of baryonic physics processes that impact the distribution of the matter (Rudd, Zentner and Kravtsov, 2008). These processes include, but not limited to, supernova feedback, active galactic nuclei feedback, and star formation. By analying hydrodynamical simulations, Daalen et al. (2011) realised that baryonic feedback affect the matter distribution in the universe at different scales, meaning that their impact should be modelled and included in a theorey of structure formation. However, the theoretical formalism of structure growth, and therefore the matter power spectrum assumes that structure growth is only driven by gravitional effects, ignoring baryonic feedback (Dodelson and Schmidt, 2026). Modelling baryonic feedback analytically is a cumbersome task, especially that the physics of galaxy formation and evolution is highly uncertain where different suits of hydrodunamical simulations have a different feedback models, and each one of these model yields a totally different

desciption of the matter power spectrum supression (Daalen et al., 2011). The mismodelling of the matter power spectrum tunnels down to the modelled cosmic shear signal and yields biased parameter estimates in a comsmological analysis.

On a conquest for precision cosmology, multiple statistical and modelling methods attempted to mititgate the impact of baryonic feedback on the large scale structure. For example, Krause et al. (2021) imposed scle cuts on measured shear correlation funciton and only considered larger scales where baryonic feedback is minimal. However, the cosmic shear signal derives most of its sensitivity from non-linear scale i.e. small scales (Piccirilli et al., 2025). Therefore, imposing a scale cut lead to loss of information. Other statistical techniques involve the use of principle component analysis to compress the parameter space of baryonic feedback parameters from hydrodynamical simulations and marginlising over these parameters in a cosmological analysis (Eifler et al., 2015). Another statistical method involves introducing an additive term to the covariance matrix called the theoretical error covariance (Moreira et al., 2021; Maraio, Hall and Taylor, 2025) which descibe the variation of different baryonic feedback scenarios among all available hydrodynamical simualtions. Analyses involving all the afformentioned results yields unbiased estimates of the cosmological parameters, however, it comes at a cost of introducing additional scatter due to the existing nuisanc parameters. Finally, A. Schneider and Teyssier (2015) introduced the baryonification correction model, a model that parametrizes the displacement of dark matter particles in the existance of baryonic feedback. This model and its developments (A. Schneider, Kovač et al., 2025) were calibrated on hydrodynamical simulations and were able to emulate baryonic feedback in other hydrodynamical simulations that they were not calibrated on. Having this model, it is possible to obtain a prediciton for any probe of baryon in the universe. In particular, this model was used to constrain baryonic feedback parameters with X-ray and the kinematic Sunyaev-Zel'dovich effect (Kovač et al., 2025) since it is possible to obtain a prediction of gas power spectra form which a simulated signal of the correponding baryon tracer could be obtained. Though these probes are promising, they rely on some assumptions with respect the structure of the galaxy clusters such as having hydrodynamic equilibrium and require many convolution from a prediciotn of the gas power spectrum up to a predicted signal. Luckily, fast radio bursts provide a direct probe of the electron density in the universe through the information that their signal encodes.

Fast radio bursts (FRBs) are pulses of electromagnetic waves of extragalactic origin. First discovered in 2007 (Lorimer et al., 2007), the signal of FRBs gets broadened while travelling through the universe, meaning that that the wave packets of higher frquency will arrive later that wave packets of higher frquency. This difference in arrival time is directly proportional to the integrated electron density along the line of sight where the dispersion measure signal could be easily simulated with minimal assumptions about the relationship of the dispersion measure and the electron number density in the universe. The grand benefit of the dispersion measure signal lies in its constraining power of baryonic feedback parameters since it is a drirect tracer of the baryon distributino in the universe. Even though the source of these bursts is unknown (Petroff, Hessels and Lorimer, 2019), it is still possible to exploit the dispersion measure signal in a cosmological context, given that FRBs are hosted within galaxies whose redshift can be identified.

Simulated likelihood, analyses showed that the cross-correlation of FRBs with other probes of structure yields better constraints on the some cosmological parameters. In particular, Calibrating baryonic feedback with FRBs and weak lesning led to one order of magnitude better constrains on the baryonic feedback and 50% improvements on constraints of the sum of neutrino masses $\sum m_\nu$ and the baryonic fraction in the unvierse Ω_b (Reischke, Neumann et al., 2023). Furthermore, combining FRBs with galaxy number counts in a 3x2-point analysis provided better constraints on baryonic feedback parameters as

compared to galaxy number counts (Sharma et al., 2026). In practice, we can aim to use FRBs alone as a probe to impose tigh constraints on the baryonic feedback parameters and use these constraints as priors in a cosmological analysis involving probes of structure, allowing the information content of LSS data to constrain the cosmological parameters and delegating the task of constraining baryonic feedback to FRBs.

The promising potential of FRBs in constraining baryonic feedback could be hindered by multiple modelling and measurement uncertainties. First, FRBs are observed using radio telescopes with only limited follow-up optical observations that are necessairy to calculate the redshifts of galaxies and by turn quantify the associated mean DM at that redshift. Second, measreuments of these redshifts are highly uncertain especially for sources at high redshifts. Third, the measured disperison measure signal receive three main contribution: a contribution from the Milkyway, a contribution from the host galaxy, and a contribution from the circugalactic medium which is the quantity of comsological interest (Reischke and Hagstotz, 2025). The models for both the Milkyway and the host galaxy contribution are both uncertain, leading to biases in the parameters constraints if not dealth with correctly. Fourth, the theoretical formalism of fast radio bursts assumes that the radio signal travels in a homogeneous universe, ignoring any gravitational lensing effect due to large scale structure. There are contradicting resutls on impact of gravitaional lensing; Alonso (2021) showed that first order contribution from general relativitstic effects to the DM angular power spectrum are subdominant compared to the contribution from the electron column density, meanwhile, Takahashi (2024) showed that the second order contribution due to lensing could lead to 1% diviation in the luminosity distance, resulting underestimation of the the luminosity distance.

Therefore, the purpose of this thesis is to model, forecast, and explore mitigation effects of systematic uncertainties and describe thrir importance with varying sample size. With upcoming radio survey, a plathora of data will be available that an accurate modelling of the dispersion measure signal is crucial to extract meanigful information from the signal (Luke Hoffmann et al., 2024). I scale up previoud work by Konar et al. (2025) by using simulation-based inference (SBI), hereafter to infer the impact of systematic uncertainties on the baryonic parameters constraints using fast radio bursts only. Using SBI in this context has multiple benefits. First, it facilitates integrating different uncertainty effects on the level the dispersion measure signal either on the level of the dispersion measure or the level of the redshift. Second, it retires the necessity to approximate the likelihood of the data given the parameters unlike traditional Baysian analyses. Results from ray tracing in hydrodynamical simualtions (Konietzka et al., 2025) suggests that a Gaussian or lognormal approximation to the probability distribution function breaks down for large number of FRBs, and analytic models of the dispersino measure is needed to simulate teh PDF of FRBs. (Torkamani et al., 2026). More details on SBI and the method used in this thesis could be found in 3.4

This thesis is structured as follows: I review the theoretical background in 2 including the general theory of relativity (2.1), the FLRW metric (2.2), the cosmological model (2.3), weak gravitational lensing (2.7), baryonic feedback on the large scale structure (2.8), and basics of fast radio bursts (2.9). I describe the methodology in 3 including modeeling the electron power spectrum (3.1), simulating the dispersion measure signal (3.2), describing the systematic unvertainties (3.3), and explainging simulation based inference (3.4). In 4, I describe the results. Finally, I dicuss the results in 5

CHAPTER 2

Theoretical Background

This chapter is mainly based on 2 textbooks and 1 lecture notes. The first book is *Modern Cosmology by* Dodelson and Schmidt (2026) and the second is *Spacetime and Geometry: An Introduction to General Relativity* by Carroll (2024). The first lecture notes comes from the *Advanced Topics in Cosmology* course given in the Summer semester 2024 by Prof. Dr. Cristiano Poriciani and Jun-Prof. Andrina Nicola. In this chapter, I will introduce the principles of cosmology, cosmic shear, and fast radio bursts (FRBs).

2.1 The general theory of relativity

The general theory of relativity introduces a geometric interpretation of gravity, resting on a set of geometric assumptions and principles evolved from the older classical Newtonian theory of gravity and relativity. The first of these is the **weak equivalence principle** which states that **' the inertial mass and gravitational mass of any object are equal'**. Einstein reformulated this statement to fit the language of the special theory of relativity by stating that **'In small enough regions of spacetime, the laws of physics reduce to those of special relativity; it is impossible to detect the existence of a gravitational field by means of local experiments.'** The consequences of this principle are that freely falling frame of reference cannot recognize that freely falling test particles in the vicinity of the frame are being accelerated (or under the influence of gravity). The consequences of this principle invites the formulation of a geometric theory of gravity where gravity manifests itself as curvature.

To have a mathematical theory where observation could be interpreted, Einstein used differentiable manifolds to describe the four dimensional spacetime. Differentiable manifolds allow a definition of physical quantities such as position and momentum along with a continuous derivative of these quantities on the manifolds. Having a differentiable quantity allows also to introduce conservation laws which are a direct results of symmetries in a particular problem (Noether, 1983). Therefore, to describe the motion of particles in a curved spacetime, the metric tensor is defined on a non-Riemannian manifold. The metric tensor is a solution for the Einstein field equations which are second order differential equations on $\mathbb{R}^{4\times4}$.

The metric tensor $g_{\nu\mu}$ (in Einstein index notation) is a 4×4 symmetric tensor that describes the physical length of an infinitesimal line given a particular system of coordinates such that

$$ds^2 = g_{\nu\mu} dx^\nu dx^\mu, \tag{2.1}$$

where $\{dx^\mu\}_{0,1,2,3}$ is a basis for the contangent bundle of the manifold. In classical terms, $\{dx^\mu\}_{0,1,2,3}$ is the differential along a coordinate axis $\{x^\mu\}_{0,1,2,3}$ For a generalized space time, the path of a free particle is described by the geodesic equation. If the coordinates of a particle in spacetime x^μ is parameterized with an affine parameter λ, then the coordinates $x^\mu(\lambda)$ is constrained by an ordinary differential equation, namely the geodesic equation given as:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma} \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0, \tag{2.2}$$

where $\Gamma^\mu_{\rho\sigma}$ are the Christoffel symbols given in terms the metric tensor such that

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2} g^{\sigma\rho} \left(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu} \right). \tag{2.3}$$

In the Newtonian limit, the gravitational potential Φ is related to the mass density ρ in the following equation

$$\nabla^2 \Phi = 4\pi G \rho, \tag{2.4}$$

with G being the gravitaional constant. To formulate a theory which reduces to Eq. 2.4, we define the energy-momentum tensor $T_{\mu\nu}$ which is a a 4×4 symmetric tensor that describes the energy density and shear in the four-dimensional spacetime. Since the gravitational field manifest itself as curvature, the analogy to Eq. 2.4 motivate a proportionality relationship between the second derivative of the metric tensor $g_{\mu\nu}$ and the stress-energy tensor $T_{\mu\nu}$

$$\left[\nabla^2 g \right]_{\mu\nu} \propto T_{\mu\nu}. \tag{2.5}$$

A geometric quantity that is composed of the second derivatives of the metric tensor is the Ricci tensor $R_{\mu\nu}$ expressed in terms of the Christoffel symbols

$$R_{\mu\nu} = \partial_\lambda \Gamma^\lambda_{\nu\mu} - \partial_\nu \Gamma^\lambda_{\lambda\mu} + \Gamma^\lambda_{\lambda\rho} \Gamma^\rho_{\nu\mu} - \Gamma^\lambda_{\nu\rho} \Gamma^\rho_{\lambda\mu}. \tag{2.6}$$

Meanwhile, the Ricci scale is defined as the contraction of the Ricci tensor

$$R = R^\mu_\mu = g^{\mu\nu} R_{\mu\nu}. \tag{2.7}$$

Therefore, a possible relationship between the stress-energy tensor and the Ricci tensor is

$$R_{\mu\nu} = \kappa T_{\mu\nu}, \tag{2.8}$$

To find κ, the conservation of energy is invoked as

$$\nabla^\mu T_{\mu\nu} = 0, \tag{2.9}$$

where ∇^μ is the covariant derivative. Following Eq. 2.8 and using Eq. 2.6 and Eq. 2.7, the covariant

derivative of the Ricci tensor is

$$\nabla^{\mu} R_{\mu\nu} = \frac{1}{2} \nabla^{\mu} R g_{\mu\nu}. \tag{2.10}$$

Therefore, we can define another tensor $G_{\mu\nu}$ with a vanishing covariant derivative

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}. \tag{2.11}$$

$G_{\mu\nu}$ is called the Einstein tensor, and the Einstein's field equation regardless

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{2.12}$$

where I used $\kappa = 8\pi G$ which can be obtained by reducing Eq. 2.12 to the Newtonian limit in Eq. 2.4. Einstein modified Eq. 2.12 to account for a vacuum solution by adding a term to represent the vacuum energy $\Lambda g_{\mu\nu}$ where Λ is called the cosmological constant, and the final equation becomes

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{2.13}$$

Finally, the 4-velocity of a particle in general relativity is defined as

$$U^{\mu} = \frac{dx^{\mu}}{d\lambda}, \tag{2.14}$$

Therefore, the 4-velocity would follow the normalization

$$g_{\mu\nu} U^{\mu} U^{\nu} = -1 \tag{2.15}$$

and the 4-momentum for a particle of mass m is defined as

$$P^{\mu} = m U^{\mu} \tag{2.16}$$

and the normalization of momentum is

$$g_{\mu\nu} P^{\mu} P^{\nu} = -m^2 \tag{2.17}$$

Notice that either of the previous normalization is invariant under coordinate transformation. We define the observed energy of a particle of 4-momentum P^{μ} with respect to an observer with 4-velocity U^{μ} such that it reduces to the special relativistic limit if the particle has a vanishing spatial velocity and the observer

$$E_{obs} = -g_{\mu\nu} U^{\mu} P^{\nu}. \tag{2.18}$$

The Einstein theory of general relativity can explain cosmological observations such as cosmological redshift and the expansion of the universe where the expansion is driven by the energy content in the universe. Though the FLRW metric (explained in 2.2) is phenomenologically motivated, the variation of the scale factor with conformal time a is described by the Einstein's field equation.

2.2 The Friedmann-Robertson-Lemaître-Walker metric

To describe the entire universe using one metric tensor requires educated simplifications and assumptions. The first guiding principle is the Copernican principle (Bondi and Roxburgh, 2010) which states that **'we do not occupy a special place in space'**. The Copernican principle motivates the cosmological principle which states that 'the universe is homogeneous and isotropic'. Early galaxy surveys have shown that the universe is only homogeneous on large scales, therefore, the full cosmological principle is stated as **'the universe is homogeneous and symmetric on large scales'**. An additional assumption that simplifies that narrows down the possible formula for a metric is Weyl's postulate which states that **'the world lines of the galaxies form a 3-sheaf of non-intersecting geodesics orthogonal to layers of spacelike hypersurfaces'**. Weyl's postulate, coupled with the cosmological principle, allows a better definition of where homogeneity and isotropy apply, namely that the universe is 'spatially homogeneous and isotropic'. Therefore, a maximally symmetric metric that follows both principles takes the form

$$ds^2 = -c^2 dt^2 + g_{ij} dx^i dx^j, \tag{2.19}$$

where t is cosmic time. A two-dimensional schematic diagram of this metric is shown in 2.1.

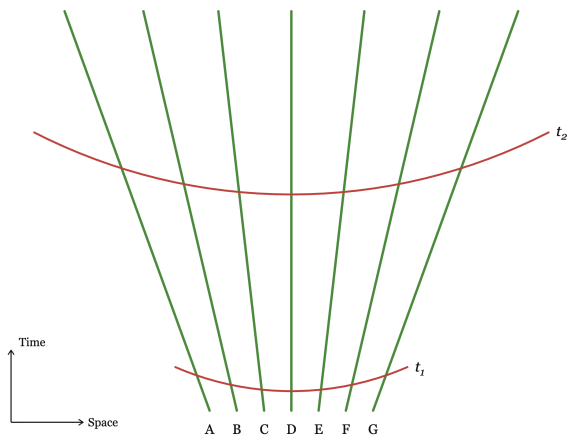

Figure 2.1: A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time t_i. These hypersuraces are homogeneous and isotropic.

Using the tools of differential geometry, a maximally symmetric metric that also accounts for a homogeneous expansion of the universe is

$$\mathrm{d}s^2 = -c\mathrm{d}t^2 + a^2(t)\mathrm{d}s_3^2 \tag{2.20}$$

where $a(t)$ is the scale factor that couples the spatial part with time and the spatial part, c is the speed of light, $\mathrm{d}s_3^2$ represent a volume element in a maximally symmetric space that takes the following form in hyperspherical coordinates

$$\mathrm{d}s_3^2 = \mathrm{d}\chi^2 + S_k(\chi)^2 \left(\mathrm{d}\theta^2 + \sin^2\theta\, \mathrm{d}\phi^2\right). \tag{2.21}$$

the function $S_k(\chi)$ takes one of three forms by virtue of the number k

$$S_k(\chi) = \begin{cases} \frac{1}{\sqrt{k}} \sin(\sqrt{k}\chi) & \text{if } k > 0, \text{for a closed universe} \\ \chi & \text{if } k = 0, \text{for a flat universe} \\ \frac{1}{\sqrt{|k|}} \sinh(\sqrt{|k|}\chi) & \text{if } k < 0, \text{for an open universe} \end{cases} \tag{2.22}$$

The case when $k = 0$ represents a flat universe, where the spatial part reduces to the usual spherical coordinates.The coordinate χ could be understood by considering two space like points in spacetime, one with $\chi = 0$ and another with $\chi = \chi_c$ leading To

$$s = a(t)\chi_c, \tag{2.23}$$

which means that the physical distance changes with time for a given χ_c. We call χ_c the comoving distance. Taking the derivative with respect to cosmic time in Eq. 2.23, we can retrieve the empirical law that Hubble (1929) discovered by recording the relationship between the distance to galaxy and the speed of the galaxy

$$\frac{ds}{dt} = V = \frac{\dot{a}}{a}s, \tag{2.24}$$

with $\dot{a} = da/dt$. We define the Hubble parameter at a cosmological time t such that

$$H(t) = \frac{\dot{a}}{a}. \tag{2.25}$$

Another distance indicator is gravitational redshift, similar to doppler redshift, gravitational redshift is the modulation of light frequency as light travels across the universe. If a fundamental observer at cosmic time t_0 emits a burst of light, then the energy of this light particle is given by Eq. 2.18. If another fundamental observer observes this light at a later time t, the ratio between the two energies is defined as

$$1 + z = \frac{E_e}{E_o}, \tag{2.26}$$

where E_e and E_o are the energy of light measured by the emitter and the observer respectively, and **z is called the cosmological redshift**. z is related to the wavelength of light in the limit where each of the emitter and the observer are fundamental observers, meaning that their 4-velocity is $U^\mu = (1, 0, 0, 0)$. Therefore, the light energy could be expressed using Planck's law (Planck, 1900), $E = h/\lambda$, with $h = 6.626 \times 10^{-34}$J·s is Planck's constant, and λ is the wavelength of light. The final relationship describing the modulation in the observed wavelength of light is

$$1 + z = \frac{a(t_o)}{a(t_e)} = \frac{\lambda_o}{\lambda_e} \tag{2.27}$$

with λ_e and λ_o are the light wavelength measured by the emitter and the observer respectively. This relationship could be exploited by quantifying the modulation of the wavelength and its relationship to the scale factor. In a nutshell, a light emitted at a scale factor $a(t_e)$ travels through an expanding universe until measured by another observer at time $t_o > t_e$ corresponding to $a(t_o) > a(t_e)$, therefore $\lambda_o > \lambda_e$, which explains why distant sources are redshifted. In practice, by measuring the spectrum of light of a distant galaxy and convolving it with a standard spectrum (the expected spectrum of galaxy in the nearby

universe), it is possible to obtain the redshift of the distant galaxy. If we choose $a = 1$ in the current epoch, the relationship between redshift and the scale factor reduces to

$$a = \frac{1}{1+z}, \tag{2.28}$$

where $z = 0$ refers to the current epoch. Since light moves along a null geodesic, it is possible to obtain the comoving distance to an emitter given the redshift of the emitter. Starting from Eq. 2.20 and Eq. 2.21, and working in a coordinate system with the light travelling along $\theta = 0$ and $\phi = 0$, the null geodesic dictates that

$$cdt = a(t)d\chi \tag{2.29}$$

which could be further arranged using Eq. 2.25 and Eq. 2.28 into

$$\chi = \int_1^{z_e} \frac{c}{H(z)} dz. \tag{2.30}$$

From the perspective of Eq. 2.28, the cosmological redshift can be used as proxy of time where larger value of redshfit conrreponds to smaller value of the scale factor a. Meanwhile from the perspective of Eq. 2.30, the cosmolgoical redshift is proxy for distance where sources of larger redshift correspond to larger distance. In a cosmological galaxy survey, measuring distances from the Earth towards galaxies is not a trivial task, however, the cosmological redshift can be measured by measuring the deviation the electromagnetic spectrum of a galaxy from the expected spectrum of a galaxy at redshift $z = 0$ [1].

Another distnace scale of interest is the Hubble scale. The Hubble scale is the distance at which the recession velocity of a galaxy equals the speed of light. Using Hubble's law, the Hubble scale is given by

$$s_H = \frac{c}{H}. \tag{2.31}$$

The Hubble scale is a distance scale that is used to define the size of the observable universe. It has a significant consequences on the growth of perturbations in the universe, as will be discussed in 2.8.

Throughout this chapter, a question is lurking behind all the kinematics of the universe that the FLRW metric proposes: how does the scale factor a change with cosmic time? In principle, this requires using the Einstein's field equation Eq. 2.13 along with assumptions about the form of the stress energy tensor $T^{\mu\nu}$. Assuming that the energy density in the universe is that of a perfect fluid, solving the Einstein's field equation for a yields

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3} \tag{2.32a}$$

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda c^2}{3} - \frac{kc^2}{a^2} \tag{2.32b}$$

with p being the isotropic pressure of the cosmic fluid and ρ being the density [2]. These equations are

[1] Of course, the process of measuring redshifts is cumbersome in practice, and this statemetn is meant to explain the basic principle behind measring the cosmological redshift.

[2] Since energy and mass are equivalent up to the squared of the speed of light, we may refer to the different constituents of the universe as 'contributing to the energy budget of the universe'.

known as the Friedmann equations, describing the evolution of the scale factor a with respect to cosmic time t. To solve these equations, a relationship between the pressure p and the energy density is needed, known as the equation of state. The possible components that contribute to the energy density ρ have been a process of trial and error starting de Sitter universe (Sitter, 1917) which is a universe with no matter up to the recently speculated model with dynamical dark energy (Adame et al., 2025). Eqs. 2.32 were derived with the assumption of a homogeneous and isotropic universe on all scales; this is not the case when studying the inhomogeneous universe with large-scale structure. Therefore, the physical quantities in Eqs. 2.32 are only a functions of time (the total density $\rho(t)$ and the isotropic pressure $p(t)$).In 2.3, I briefly present the case of inhomogeneous universe with perturbations

$$\frac{\rho(t,\mathbf{x}) - \bar{\rho}(t)}{\bar{\rho}(t)} << 1, \tag{2.33}$$

where $\bar{\rho}(t)$ is the average density at a cosmic time t, and $\rho(t,\mathbf{x})$ is the density at a comoving position $\mathbf{x}$ and cosmic time t. In this thesis, we consider the ΛCDM cosmological model and describe its components in 2.3.

2.3 The cosmological model

In this section, we consider the ΛCDM cosmological model and explain its components. As noted by Eqs. 2.32, the time evolution of the scale factor a depends on the density and its relation to the pressure, known as the equation of state. If we refer to the current day value of the Hubble parameter as H_0 and the current day value of the density ρ_0, Eq. 2.32b can be written in the current epoch

$$1 = \frac{8\pi G}{3H_0^2}\rho_0 + \frac{\Lambda c^2}{3H_0^2} - \frac{kc^2}{H_0^2}. \tag{2.34}$$

Notice that the quantitiy $\frac{3H_0^2}{8\pi G}$ has the dimensionality of a density, and we call it the critical density ρ_{cr}. It is possible to express the density fractions of the different components in terms of the current day value of the component's density and the critical density.

$$\Omega_0 = \frac{\rho_0}{\rho_{cr}}, \tag{2.35a}$$

$$\Omega_\Lambda = \frac{\Lambda c^2}{3H_0^2}, \tag{2.35b}$$

$$\Omega_k = -\frac{kc^2}{H_0^2}. \tag{2.35c}$$

We list below the constituents of the ΛCDM model of cosmology. It is common to express the density of a constituent as a fraction of the critical density, such that the sum of these fractions equals 1. In practice, the density fraction of a constituent X in the current epoch Ω_X is expressed as

$$\Omega_X = \frac{\rho_{0,X}}{\rho_{cr}}, \tag{2.36}$$

with $\rho_{0,X}$ being the density of a constituent X at the current epoch.

Measurements from the cosmic microwave background has shown that we live in a flat universe (Smoot, 1999), meaning that $\Omega_k = 0$ (within error bounds). **For the rest of this work, I assume a flat universe**. The two remaining components are the density fraction Ω_0 and the fraction Ω_Λ. The mechanism of how these components were formed into existence will be omitted from this thesis. It is worth noting that the Hubble parameter H_0 is conventionally given in term of a dimensionless number h such that

$$H_0 = 100\, h\, \mathrm{km\, s^{-1}\, Mpc^{-1}}, \tag{2.37}$$

and many other quantities are expressed in terms of h; for example, the density ratios Ω_X could be given as $\Omega_X h^2$. Measurements of the CMB with the Planck satellite (Aghanim et al., 2020) reported a result of

$$H_0 = 67.66 \pm 0.42 \mathrm{km\, s^{-1}\, Mpc^{-1}},$$

meanwhile measurements using supernovae from the SH0ES collaboration (A. G. Riess, Yuan et al., 2022) report a value of

$$H_0 = 73.04 \pm 1.04 \mathrm{km\, s^{-1}\, Mpc^{-1}}.$$

The discrepancy of H_0 measurements between the early universe with the CMB and late universe measurements with supernovae is called the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024).

The equation of state of each component in the ΛCDM model differs between the early universe with large energy scales and the late universe with very low energy scales. We list below each component and its role in the cosmic evolution of the universe. In general, the equation of state $p_X(\rho_X)$ expresses the relationship between pressure p_X and density ρ_X of a constituent X. Motivated by dimensional analysis [3], we can conclude that the pressure and density are related

$$p_X = w \rho_X c^2. \tag{2.38}$$

The relationship between the scale factor a and the density ρ_X could be found by invoking the first law of thermodynamics and solving the following ordinary differential equation,

$$\mathrm{d}(\rho_X c^2 a^3) = -p_X\, \mathrm{d}(a^3). \tag{2.39}$$

In total, the ΛCDM universe contains the following constituents:

- **Radiation:** Radiation is mostly composed of freely propagating relativistic particles: photons and neutrinos. Photons have been propagating in the universe since recombination. Recombination is the epoch when free electrons combine with nuclei to form neutral atoms, after which the photon mean free path becomes effectively infinite and photons decouple from matter; these photons are observed today as the cosmic microwave background (CMB) radiation. CMB anisotropies contain a wealth of information about the density fractions of the other components of the Universe and its spatial curvature. The temperature of the photon background scales as $T_r \propto a^{-1}$. With equation of state

$$p_r = \frac{1}{3} \rho_r c^2 \tag{2.40}$$

[3] Dimensional analysis is the method of comparing the units of two physical quantities to conclude the relationship between them up to a proportionality constant.

and the radiation energy density scales as $\rho_r \propto a^{-4}$.

$$\rho_r = \rho_{r,0} a^{-4} \tag{2.41}$$

In addition to photon, neutrinos contribute to the radiation budget in the universe. At early times, neutrinos are in equilibrium with other particles such as electrons and photons, however, as the universe expands and the temperature drops, electrons no longer interact with neutrinos, therefore, neutrinos decouple from the rest of the components in the universe in a process known as freeze-out. The density of neutrinos scales to the scale factor in a similar way to photons, however, the present day temperature of neutrinos is different from that of photons

- **Baryonic matter:** Baryonic matter is composed of the particles that make up the standard model of particle physics. The particles that make up this component interact with light and with itself through long range pressure. In the early universe, baryonic matter is coupled to photons through absorption, emission, and scattering processes. Due to the expansion of the universe, the absorption cross-section approaches zero, therefore, the photons emitted from recombining electrons with nuclei have less energy than needed to ionize a similar atom, leaving the photon free. The process of forming neutral atoms by combining free electrons with nuclei is called recombination. This process continues until all the atoms are neutral, and then, photons propagate freely in the universe, with a very low probability of interacting with Baryons. After an epoch of structure formation (with the universe being transparent), neutral gas clouds in dark matter halos collapse under gravity to form stars and trigger fusion reactions [4]. Photons emitted from stars reionize the medium around stars, and baryons interact again with photons (Draine, 2011). The fraction of baryonic matter in the universe is measured to be $\Omega_b h^2 = 0.0225 \pm 0.0003$. In the early universe, the equation of state of baryonic matter is derived with relativistic consideration, however, after recombination, the short range pressure of baryons is negligible and the equation of state for baryons is that of dark matter on large scales. On small scale, baryonic processes in the late universe are highly non-linear and cumbersome to model which impose a limitation on the modelling of the cosmic shear signal. **This is the exact problem that this thesis is trying to tackle**.

- **Dark matter:** Dark matter is matter that does not interact with light, however, its effect is observed gravitationally. Recent results from different cosmological probes including CMB, galaxy clustering and weak lensing favour a cold dark matter component, meaning that the temperature of dark matter is much less than its mass. As a consequence, dark matter could be modelled as a pressureless fluid on large scale (on small scales, the fluid approximation breaks down). Using this simple picture of cold dark matter (CDM), it is possible to solve the fluid equation of dark matter to linear order and obtain a good approximation of the linear density evolution with cosmic time (I show this derivation below). The equation of state of dark matter, therefore, is simply $p = 0$. By solving Eq. 2.39 for dark matter, the density of dark matter depends on the scale factor a such that

$$\rho_c = \rho_{c,0} a^{-3} \tag{2.42}$$

The present day value of the dark matter density fraction is measured to be $\Omega_m h^2 = 0.1431 \pm 0.0025$

[4] A fusion reaction is a nuclear reaction when two low mass atoms combine to form a heavier atom, emitting an amount of energy.

by the Planck collaboration (Aghanim et al., 2020). Dark matter is the main driver of matter clustering in the universe since dark matter can form virialized region of space called dark matter halos which accretes baryons and leads to galaxy formation as a consequence (Wechsler and Tinker, 2018).

- **Dark energy:** dark energy is a term coined by Michael S. Turner (Turner, 2001) to describe the contribution to the energy budget that is not in a form of particles. From Eqs. 2.35, dark energy is the component with density fraction Ω_Λ. Early on, this component was considered to be smooth and homogeneous, meaning that its corresponding density $\rho_\Lambda = \Omega_\Lambda \rho_{cr}$ only varies with cosmic time. In its simplest form, the dark energy pressure is related to the dark energy density as follows

$$p_\Lambda = w_\Lambda \rho_\Lambda c^2 \tag{2.43}$$

with the simplest case $w_\Lambda = -1$. The equation of state of dark energy is highly debated with many modification suggested to explain tensions in cosmology such as the Hubble tension (Verde, Schöneberg and Gil-Marín, 2024). In this work, we assume an equation of state of dark energy with $w_\Lambda = -1$. The Planck collaboration (Aghanim et al., 2020) reported that $\Omega_\Lambda = 0.6889 \pm 0.0056$.

Figure 2.2 shows a schematic representation of the components discussed above. In general, the matter density fraction Ω_m is the sum of the baryonic matter density fraction Ω_b and the dark matter density fraction Ω_c. The total energy budget of the universe is the sum of the matter density fraction Ω_m, the dark energy density fraction Ω_Λ, and the radiation density fraction Ω_r. Since the radiation density fraction is negligible in the current epoch, we have $\Omega_m + \Omega_\Lambda \approx 1$. With that, the evolution of the expansion rate of the universe in terms of the scale factor a is given by

$$H^2(a) = H_0^2 \left[\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda \right], \tag{2.44}$$

where the different additive terms dominate the evolution at different epochs. In particular, dark energy is the dominant component in the current epoch. To constrain the equation of state of dark energy, cosmic shear stands as a powerful probe of late structure formation since it is an integrated qunatity that includes signficant contribution from the late evolution of the universe, and therefore sensitive to dark energy. The basic principles of cosmic shear and its relationship to the matter density field are explained in **??**. This thesis focuses on using FRBs to constain baryonic feedback on the large scale structure and therefore improve the modelling of cosmic shear on small scales, which in turn improves the constraining power of cosmic shear on dark energy.

So far, I have only introduced the universe as a homogeneous entity, ignoring the fact the universe is inhomogeneous on small scales. As a physical system, the initial conditions of the universe are scale-dependent Gaussian perturbations which grow to be highly non-linear (and non-Gaussian) large perturbations (large relative to the mean density of the universe). Since the initial perturbations were generated via a stochastic process, the resultant perturbations are called random fields. In particular, the matter density field $\rho_m(\mathbf{x}, t)$ is the random field that describes the matter density at a comoving position $\mathbf{x}$ and cosmic time t. For simplicity, the matter perturbations are often expressed in terms of the matter density contrast δ_m such that

$$\delta_m(\mathbf{x}, t) = \frac{\rho_m(\mathbf{x}, t) - \bar{\rho}_m(t)}{\bar{\rho}_m(t)}, \tag{2.45}$$

where $\bar{\rho}_m(t)$ is the average matter density at a cosmic time t. In addition, similar definitions apply to

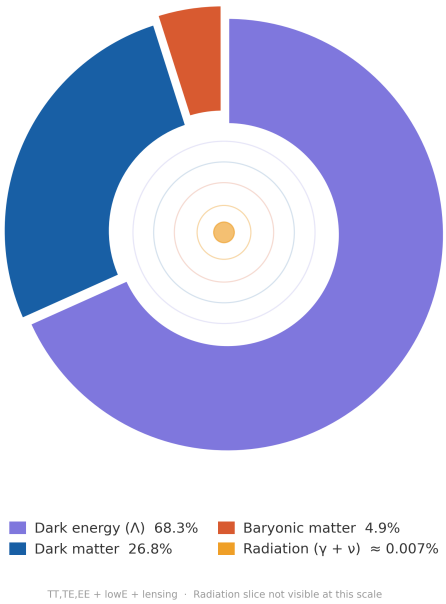

Figure 2.2: A schematic representation of density fractions of the different components in the ΛCDM model. The density fractions are measured by the Planck collaboration (Aghanim et al., 2020) using measurements of the CMB anisotropies.

any other density field such as the baryonic density contrast or the electron density contrast, which will be important in the study of FRBs. By construction, any density contrast field is a random field with a mean of zero i.e. $\langle \delta_m \rangle = 0$. Higher order statistics of a random field also encode information about the random field. This thesis only uses the two-point statistics of the matter density contrast and its Fourier space counterpart, namely the power spectrum. We discuss these statistics below. Since dark matter is the dominant component of matter in the universe, it is the component responsible for the observed structure in the universe. In addition, the pressure induced by baryonic processes is negligible on large scales, meaning that the evolution of matter density contrast on large scales is well approximated by the evolution of dark matter density contrast. Therefore, the evolution of matter density contrast δ_m is well approximated by the evolution of dark matter density contrast δ_c. I present below the Einstein-Boltzmann equation that describes the phase-space evolution of a particle species in the universe.

2.4 The Boltzmann equation & the linear perturbation theory

This section is partly based on a review on the effective field theory of large scale structure (EFTofLSS) by Tobias Baldauf (Baldauf, 2020). The Boltzmann equation describes the evolution of the phase-space distribution function of a particle species in the universe. This function is labelled as $f(\mathbf{x}, \mathbf{p}, t)$, where $\mathbf{x}$ is the position, $\mathbf{p}$ is the momentum, and t is the cosmic time. Different observables could be obtained from the distribution function f including the density $\rho(\mathbf{x}, t)$ and the velocity field $\mathbf{v}(\mathbf{x}, t)$. The Boltzmann equation for a collisionless and pressureless dark matter

$$\begin{aligned}\frac{\mathrm{d}f}{\mathrm{d}t} &= \frac{\partial f}{\partial t} + \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}\cdot\frac{\partial f}{\partial \mathbf{x}} + \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}\frac{\partial f}{\partial \mathbf{p}} \\ &= \frac{\partial f}{\partial t} + \frac{\mathbf{p}}{ma}\cdot\frac{\partial f}{\partial \mathbf{x}} - am\nabla\phi\cdot\frac{\partial f}{\partial \mathbf{p}} = 0,\end{aligned} \tag{2.46}$$

with m being the mass of the dark matter particle. If dark matter behaves as a fluid, the density and the bulk velocity of the fluid could be obtained by taking the first two moments of the distribution function f such that

$$\rho_m(\mathbf{x},t) = \frac{m}{a^3}\int \mathrm{d}^3p\, f(\mathbf{x},\mathbf{p},t), \tag{2.47}$$

$$v_i(\mathbf{x},t) = \int \mathrm{d}^3p\, \frac{p_i}{am} f(\mathbf{x},\mathbf{p},t) \Bigg/ \int \mathrm{d}^3p\, f(\mathbf{x},\mathbf{p},t), \tag{2.48}$$

$$\sigma_{ij}(\mathbf{x},t) = \int \mathrm{d}^3p\, \frac{p_i}{am}\frac{p_j}{am} f(\mathbf{x},\mathbf{p},t) \Bigg/ \int \mathrm{d}^3p\, f(\mathbf{x},\mathbf{p},t) - v_i(\mathbf{x})v_j(\mathbf{x}). \tag{2.49}$$

with $\sigma_{ij}(\mathbf{x},t)$ representing the velocity dispersion tensor. The distribution function f has higher order moments, however, the anisotropic stress tensor is second order in momentum, therefore, it can be postulated that it is negligible at linear order and at early times where the velocity dispersion is small[5]. In this thesis, I am only interested in the evolution of the matter density contrast. Using Eq. 2.46, the evolution of the matter density contrast and the bulk velocity are governed by the continuity and the Euler equations. The equations up to linear order are as follows

$$\frac{\partial \delta_m}{\partial \tau} + \frac{1}{a}\nabla\cdot\mathbf{v} = 0, \tag{2.50a}$$

$$\frac{\partial \mathbf{v}}{\partial \tau} + \frac{1}{a}(\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{1}{a}\nabla\phi, \tag{2.50b}$$

with τ being the conformal time defined as $\mathrm{d}\tau = \mathrm{d}t/a$. Eqs. 2.50 show that at a linear order the matter density contrast is coupled to the velocity field, and the velocity field is coupled to the gravitational field ϕ. The growth of structure is, therefore, sourced by gravity.

To solve the equations in real space is a cumbersome task, therefore, we take the following steps in order to obtain the evolution of the matter density contrast.

- Apply the Fourier transform to Eqs. 2.50 to obtain the equations in Fourier space. The operator ∇ reduces to a multiplication by $i\mathbf{k}$, with $\mathbf{k}$ being the Fourier space wavevector.
- The velocity field is a vector field, therefore, it can be decomposed into a curl-free component and a divergence-free component. The curl-free component is called the velocity divergence θ and is defined as $\theta = \nabla\cdot\mathbf{v}$, while the divergence-free component is called the vorticity ω and is defined as $\omega = \nabla\times\mathbf{v}$. At linear order, the vorticity decays with time, therefore, it can be neglected at linear order.

[5] The usual practice in standard perturbation theory is to truncate the Boltzmann hierarchy even for higher order perturbations

- The peculiar gravitational potential ϕ can be related to the matter density contrast by the relationship known as the Poisson equation such that

$$\nabla^2\phi = \frac{3}{2}\Omega_{\rm m}H_0^2\frac{\delta_m}{a}. \tag{2.51}$$

With these steps, the equation that governs the eveolution of the matter denstiy field at a linear level is a second order ordinary differential equation with respect to the conformal time τ such that

$$\frac{{\rm d}^2\delta_m}{{\rm d}\tau^2} + \mathcal{H}\frac{{\rm d}\delta_m}{{\rm d}\tau} - \frac{3}{2}\Omega_{\rm m}H_0^2\frac{\delta_m}{a} = 0, \tag{2.52}$$

with $\mathcal{H} = aH$ being the conformal Hubble parameter. A solution for the matter density contrast δ[6] at linear order in Fourier space as follows

$$\delta(\mathbf{k}, t) = D_+(\tau)\delta_{+,0}(\mathbf{k}), \tag{2.53}$$

where $\delta_{+,0}$ is the initial conditions of the matter density contrast, and $D_+(\tau)$ is the growth factor whose exact expression depends on the cosmological model considered. For an Einstein-de Sitter universe, which is a universe with $\Omega_m = 1$ and $\Omega_\Lambda = 0$, the growth factor is simply $D_+(\tau) = a(\tau)$. The importance of the result 2.53 lies in two observations: the first is that the growth of each Fourier mode of the matter density contrast is independent from other modes (modes of different length scales are not coupled), and the second is that the time evolution of the matter density contrast is separable from its spatial dependence.

The solution to Eqs. 2.50 is given assuming that the universe is way into matter dominated epoch, where the impact of radiation is negligible. In the early universe, the radiation component is dominant and initiates pressure that only acts on sacles smaller that the horizon scale, defined in Eq. 2.31, since pressure is a causal process that can only propagate at a speed less than the speed of light. Therefore, $\delta_{+,0}(\mathbf{k})$ should encode all the physics of the early universe including the impact of radiation on the matter density contrast and the initial conditions. In practice, comparing the evolved density field with observaiton will not give a conclusive answer on whether the theory presented here is correct or not, since the current universe is one realisaiton of the stochastic process that generated the initial conditions, therefore, estimating the statistical properties of the observed matter density field and comparing it with the theoretical prediction is the way to test the theory. In the next section, I present the power spectrum as a statistic that encodes the statistical properties of the matter density field.

At early times, the amplitude of the matter density contrast is small, therefore, the linear solution in Eq. 2.53 is a good approximation to the evolution of the matter density contrast. As time goes on, the amplitude of the matter density contrast grows and the linear approximation breaks down. Furthermore, for regions in the universe where the average density within this specfic region is large compared to the average density of the universe, dark matter paricles collapse under the influence of gravity and virialised regions of space form. These virialised regions are called dark matter halos. The appendix include an explanation of the toy model for the collapse. add the number of the appendix. Baryons fall in the deep potential well of the dark matter halo and accretes to form galaxies, meaning that galaxies traces the position of dark matter halos, and in turn dark matter.

[6] δ refers to the matter density contrast δ_m hereafter, and the subsript is omitted for ease of writing.

2.5 The matter power spectrum and dark matter tracers

Since $\delta_0(\mathbf{k})$ is a Gaussian random field, its statistical properties are completely determined by its mean and variance. The mean of $\delta_0(\mathbf{k})$ is zero by construction, therefore, the variance is the only quantity that encodes the statistical properties of $\delta_0(\mathbf{k})$. The variance of $\delta_0(\mathbf{k})$ is called the power spectrum and is defined as

$$\langle \delta_0(\mathbf{k})\delta_0^*(\mathbf{k}')\rangle = (2\pi)^3 P(k)\delta_D(\mathbf{k}-\mathbf{k}'), \tag{2.54}$$

where δ_D is the Dirac delta function. Notice that the power spectrum is only a function of the magnitude of the Fourier space wavevector $k = |\mathbf{k}|$ due to the statistical isotropy of the universe. Fig. 2.4 shows the matter power spectrum obtain using the CLASS Boltzmann solver (Lesgourgues, 2011) with varying fraction of baryonic matter Ω_b and varying Hubble parameter H_0 at redshift $z = 0$.

Though the matter power spectrum is a powerful statistic that encodes the statistical properties of the matter density field, it is not directly observable. In order to relate the matter field to obervables, it is important to understand the physical mechanisms by which the observables are related to the matter field. For galaxies, galaxy formation theories suggest that baryonic matter falls into the potential well of a dark matter halo and accretes to form a galaxy. Fig. 2.3 illustrastes a snapshot of hydrodynamical simualtion with a dark matter halo hosting a galaxy.

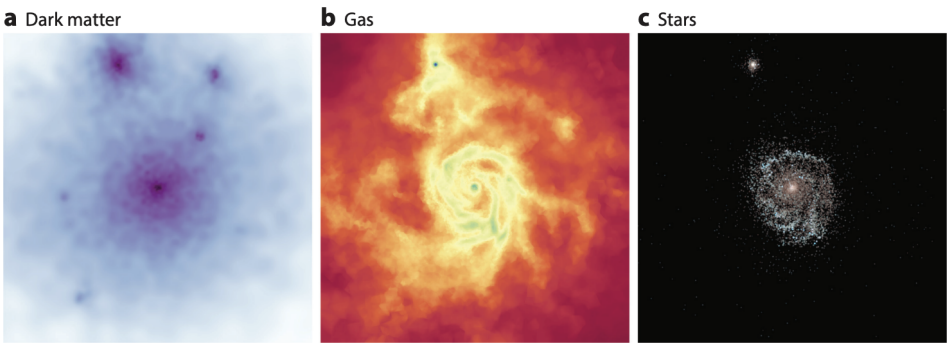

Figure 2.3: A 2D plot from the IllustrisTNG simulation (Pillepich et al., 2018), adopted from Crain and Voort (2023). It is a snapshot of hydrodynamical simulation showing how a 'danser than average' region of dark matter hosts a galaxy (stars make up the 'skeleton' of a galaxy).

The detailed mechanism of how this happens is highly non-linear and complicated. In addition, running comprehensive galaxy formation simulations is computationally expernsive in order to cover the large parameter space of galaxy formation physics (Somerville and Davé, 2015). Therefore, it is common to use a phenomenological model to relate the galaxy field to the matter field. The simplest model is the linear bias model which assumes that the galaxy number density contrast δ_g is linearly related to the matter density contrast δ such that

$$\delta_g(\mathbf{x}, t) = b(t)\delta(\mathbf{x}, t), \tag{2.55}$$

with b being the linear bias parameter. The linear bias model is a good approximation on large scales, however, on small scales, the relationship between the galaxy field and the matter field is highly non-linear and the linear bias model breaks down (Desjacques, Jeong and Schmidt, 2018). Notice that the galaxy number density contrast is only dependent on the local matter density field and does not depend on

the environment surrounding the galaxy. This approximation breaks down on small scales where th environment of a galaxy can have a significant impact of the galaxy formation process. Galaxies are called 'biased tracers of dark matter' since they trace the underlying matter field.

It is possible to use the same formalism of galaxy bias to relate the matter density contrast with other tracers of dark matter such as dark matter halos. In practice, the bias of dark matter halos is related to the underlzing matter field via the following relationship

$$\delta_h(\mathbf{x}, t) = b_h(t)\delta(\mathbf{x}, t), \tag{2.56}$$

with b_h being the dark matter halo bias parameter. Owing to Eq. 2.56, the halo power psectrum can be expressed in terms of the matter power spectrum defined in Eq. 2.54 such that

$$P_{hh}(k, t) = b_h^2(t)P_{mm}(k, t) \tag{2.57}$$

For dark matter halos, it is possible to go bayond a linear bias model by fitting an empirical model of halo bias to the results of an N-body gravitional simulation. In this thesis, I use the fitting formula of Tinker et al. (2010) to model the halo bias. In this recipe, the halo bias is a function of the halo mass and its redshift $b_h(M, z)$. As a consequence, the halo power spectrum is related to the matter power spectrum with the relationship

$$P_{\mathrm{hh}}(M_1, M_2, k) = b(M_1)b(M_2)P_{\mathrm{mm}}^{\mathrm{lin}}(k)[1 + \beta^{\mathrm{nl}}(M_1, M_2, k)] \tag{2.58}$$

In the next section, I present the halo model, a non-pertubative approximation to obtain the non-linear power spectrum of matter or a tracer of matter. In particular, this thesis is concerned with the electron power spectrum in order to model the cosmological contribution of the dispersion measure of fast radio bursts.

2.6 The halo model & the electron power spectrum

The details in this section are based on a review of the halo model by Asgari, Mead and Heymans (2023). The halo model assumes that matter in the universe are entirely contained in dark matter halos. Therfore, any cosmological field δ_{u} at a comoving coordinate $\mathbf{x}$ can be expressed in terms of the halo denisty profile $W_{\mathrm{u},i}$ such that [7]

$$\delta_{\mathrm{u}}(\mathbf{x}) = \sum_i N_i\, W_{\mathrm{u},i}\left(\left|\mathbf{x} - \mathbf{x}_i\right|\right), \tag{2.59}$$

where the sum runs over all volume elements and $N_i = \{0, 1\}$ determines whether there is a halo centre in that volume element. $W_{\mathrm{u},i}$ could express the halo density profile for any comp onent that traces halos such as electrons. In N-body simulations, the halo profile $W_{\mathrm{u},i}$ of a particular dark matter halo i difffers from one halo to another as a funciton of the halo mass M. To express this scatter, the halo-mass distribution function is introduced. The halo mass distrinution function $n(M)$ is defined as the number of dark matter halos with $M_i \in [M, M + \mathrm{d}M]$ is $n(M)\mathrm{d}M$. Averaging over all possible halo-masses, the expectation value of a field becomes

[7] notice that the time dependency is omitted for ease of writing.

$$\langle \delta_{\rm u}(\mathbf{x})\rangle = \left\langle \sum_i N_i\, W_{\rm u}(M_i, |\mathbf{x}-\mathbf{x}_i|)\right\rangle = \int_0^\infty {\rm d}M\, n(M) \int {\rm d}^3x'\, W_{\rm u}(M, |\mathbf{x}-\mathbf{x}'|) = \int_0^\infty {\rm d}M\, W_{\rm u}(M) n(M)\,. \tag{2.60}$$

where $W_{\rm u}(M, \mathbf{x})$ can be split into two multiplicative component: $W_u(M)$ which encodes the amplitud eof the halo profile and $U_{\rm u}(M, \mathbf{x})$ whichc encodes information about the shape of the profile, andd the mass-dependent halo profile can be written as

$$W_{\rm u}(M, x) = W_{\rm u}(M) U_{\rm u}(M, x), \tag{2.61}$$

with $W_{\rm u}(M)$ appearing in the last equality, and $U_{\rm u}(M, x)$ is regourously defined as

$$\int {\rm d}^3x\, U_{\rm u}(M, x) = 1\,. \tag{2.62}$$

The physical intuition behind the function $W_{\rm u}(M, x)$ can be understood by going to Fourier space. First, Eq. 2.62 indicates that the Fourier space counterpart of the function $U_{\rm u}(M, x)$, namely $\hat{U}_{\rm u}(M, k)$ has the property $\hat{U}_{\rm u}(M, k \to 0) = 1$ which implies that $\hat{W}_{\rm u}(M, k \to 0) = \hat{W}_{\rm u}(M)$. THis result is understood that at large scales, dark matter halos act as point masses (similar to a Dirac-delta function) which translates to a constant in Fourier space.

With this formalism in place, the ingredients of the halo model allows an esitmation for the power spectrum of a species u (in the case of this thesis, the halo model formalism will be applied to obtain a prediction of the electron density in the universe). Using the definition of the pwoer spectrum in Eq. 2.54, the power spectrum of the field δ_u as described inthe halo model is

$$\langle \hat{\delta}_{\rm u}(\mathbf{k}) \hat{\delta}^*_{\rm u}(\mathbf{k}')\rangle = \left\langle \sum_{i,j} e^{-i\mathbf{k}\cdot\mathbf{x}_i}\, e^{i\mathbf{k}'\cdot\mathbf{x}_j}\, N_i\, N_j\, \hat{W}_{{\rm u},i}(M, k)\, \hat{W}_{{\rm u},j}(M, k')\right\rangle. \tag{2.63}$$

The sum can be further divided into two contributions:

- a contribution with $i = j$ which is the power restultant from the correlation of dark matter particles within one dark matter halo given by

$$P^{\rm 1h}_{\rm uu}(k) = \int_0^\infty \hat{W}_{\rm u}(M, k)\, \hat{W}_{\rm u}(M, k)\, n(M)\, {\rm d}M\,, \tag{2.64}$$

  and this contribution is termed as **the one-halo term**.

- a contrubution with $i \neq j$ whcih is the power resultant from the correlation of dark matter particles in two different halos given as

$$P^{\rm 2h}_{\rm uu}(k) = \int_0^\infty \int_0^\infty {\rm d}M_1\, {\rm d}M_2\, P_{\rm hh}(M_1, M_2, k)\, \hat{W}_{\rm u}(M_1, k)\, \hat{W}_{\rm u}(M_2, k)\, n(M_1)\, n(M_2), \tag{2.65}$$

  and this contribution is termed as **the two-halo contribution**. P_{hh} is the halo power spectrum defined in Eq. 2.58.

The non-linear power spetrum wihtin the halo model of a species u is given by

$$P_{uu}(k) = P^{1h}_{uu}(k) + P^{2h}_{uu}(k) \tag{2.66}$$

Fig. 2.4 shows an example of the non-linear power specrutm with the different contributions from the one-halo and the two-halo terms. Since P^{1h}_{uu} describes the **inter-halo** clustering of a species u, it dominates the power spctrum on small scales. Meanwhile,

Naturally, the halo-model is a recipe for obtainting an approximnation of the non-linear pwoer spectrum. By knowing the halo profile of a spcies u, it is possible to obtain the non-linear power spectrum of this species. In this thesis, we use the halo model to obatin a theoretical prediction of the electron power spectrum. Section 3.1 describes the procedure of obtaining the electron power spectrum using the baryonfication correction model, and section 2.8 describes how electron profiles are obtained usinga model for baryonic feedback.

Nonetheless, the halo model has its shortcoming (Mead et al., 2021). The power spectrum estimation via the halo model agrees with the power specrtum estimation from N-body gravitional simualtions on very large and very small scales. Meanwhile, it fails on intermediate scales, the scales where the transition from the two- to the one-halo term. However, for the use case specific to this thesis, the halo model is useful since it is able to model the small scales which are mostly impacted by baryoinic feedback accurately.

2.7 Weak gravitational Lensing & cosmic shear

This section is based on the lecture notes of the *Gravitational Lensing* cousrse taught by Prof. Dr. Peter Schneider at the University of Bonn.

Gravitational lensing, in the language of general relativity, is the modification of light (null) geodesics by virtue of the existance of massive objects. In the language of special relativity, it is the bening of light by virtute of the gravitional force of a massive object on a photon. As a result, the image of a bright object (galaxy or a quasar for example) gets magnified and sheared as compared to a situation where there are no lensing effects. In this thesis, I am only interested with the weak field limit of the gravitional lensing where only shear effects are observable.

2.8 The impact of baryonic feedback on the large-scale structure

results from hydrodynamical simulations, cosmology analyses resutls. show the power spectrum supression. Speak about how different scenarios of baryonic feedback lead to different levels of power spectrum suppression. Show how the power spectrum suppression leads to a bias in cosmological parameters if not modelled correctly.

2.9 Fast radio bursts

- show the relationship between the ispersion measure and the time of arrival.
- Speak about hte collaborations to find FRBs.
-

describe how modelling the PDF is a hard thing to do To forward model observations, we convolve the $DM - z$ represented by $P(DM|z)$ with measurement and systematic uncertainties, in addition to other observational effects. A catalog of detected FRBs is expected to have observed dispersion measure $DM^i_{\rm obs}$, galactic coordinates, and redshifts for a subset of and observations z^i. We assume that each detected FRB was emitted at redshift z^i where the difference between identified and unidentified host can be integrated in how $P(z^i)$ is modeled. Assume that we observe N_{FRB} fast radio bursts signal. We express the observations as the probability of detecting an FRB $DM_{\rm obs}$ given a set of estimated redshifts $\{z^i\}$, can be decomposed using the Bayes rule such that

$$P(DM_{\rm obs}|\{z^i\}) \propto P(\{z^i\}|DM_{\rm obs})P(DM_{\rm obs}) \tag{2.67}$$

Furthermore, if we assume that each observation z_i is independent of other observations we may use the Bayes rule to decompose $P(\{z^i\}|DM_{\rm obs})$ such that

$$P(\{z^i\}|DM_{\rm obs}) = \prod_i P(z^i|DM_{\rm obs}) \tag{2.68}$$

Furthermore, the individual terms in Eq. 2.68 could be derived using the joint probability distribution of the observed and the dispersion measure contribution from the large scale structure, namely $DM_{\rm obs}$ and DM such that

$$\begin{aligned} P(z^i|DM_{\rm obs}) &\propto P(DM_{\rm obs}|z^i)P(z^i) \\ &= P(z^i)\int P(DM_{\rm obs}, DM|z^i)dDM \\ &= P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.69}$$

The final expression that relates the probability distribution function of an observation to other modelled terms is

$$\begin{aligned} &P(DM_{\rm obs}|\{z^i\}) \propto \\ &\qquad P(DM_{\rm obs})\prod_i P(z^i)\int P(DM_{\rm obs}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.70}$$

Furthermore, to model each component of the resulting expression, we start with $P(DM_{\rm obs})$. There are three components that contribute additively to $DM_{\rm obs}$: the large scale structure contribution $DM_{\rm LSS}$, the Milkyway contribution $DM_{\rm MW}$, and the host galaxy contribution $DM_{\rm host}$, and the observed dispersion measure is given as

$$DM_{\rm obs} = DM_{\rm LSS} + DM_{\rm MW} + DM_{\rm host} \tag{2.71}$$

For simplicity, we indicate to the contribution from the large-scale structure simply with DM which is the same in Equation Eq. 2.70. Consider the joint probability distribution $P(DM, DM_{\rm MW}, DM_{\rm host})$, this can be related to $P(DM_{\rm obs}) = P(DM + DM_{\rm MW} + DM_{\rm host})$ via the constrained integral

$$
\begin{aligned}
P(DM_{\text{obs}}) &= P(DM + DM_{\text{MW}} + DM_{\text{host}}) \\
&= \int_C P(DM, DM_{\text{MW}}, DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}} \\
&= \int_C P(DM)P(DM_{\text{MW}})P(DM_{\text{host}})\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}}
\end{aligned}
\tag{2.72}
$$

Where $C = \{DM, DM_{\text{MW}}, DM_{\text{host}}; DM_{\text{obs}} = DM + DM_{\text{MW}} + DM_{\text{host}}\}$ Similarly, I can model $P(DM_{\text{obs}}|z^i, DM)$ as

$$
\begin{aligned}
P(DM_{\text{obs}}|z^i, DM) = \int_C & P(DM|z^i, DM)P(DM_{\text{MW}}|z^i, DM) \\
& P(DM_{\text{host}}|z^i, DM)\, \mathrm{d}DM\, \mathrm{d}DM_{\text{MW}}\, \mathrm{d}DM_{\text{obs}}
\end{aligned}
\tag{2.73}
$$

Each of the terms making up the integral could be further reduced

$$P(DM|z^i, DM) \propto P(DM|z^i)P(DM) \tag{2.74}$$

$$P(DM_{\text{MW}}|z^i, DM) = P(DM_{\text{MW}}) \tag{2.75}$$

$$P(DM_{\text{host}}|z^i, DM) \propto P(DM_{\text{host}}|z^i)P(DM|z^i) \tag{2.76}$$

Keeping in mind, that DM_{host} that we use in Eq. 2.70 is flat since we do not have any information of the

CHAPTER 3

Methodology

In this section, we explain the methoddology used in this thesis.

Add a flow chart that summarizes the different steps in the pipeline, like the one in the interview presentation.

Add a table showing the cosmological parameters.

Consider adding a table which describes the fiducial cosmology.

3.1 Modelling the electron power spectrum

explain things in a schematic representation first.

3.1.1 The baryonification correction model

explain how we ignore the redshift evolution of some parameters.

3.1.2 The halo model

Check that you have explained what dark matter halos are in the beginning. Refer to Either the review by Marika or the paper on the Web halo model. Mention that we use the Tinker halo mass function and the NFW profile.

The Core Cosmology Library (CCL)

3.2 Simulating the dispersion measure signal

Add a table with the fiducial values and their prior. explain how we ignore the redshift evolution of some parameters. Motivate the choice for the number of shells and the redshift range.

3.2.1 The Generator for Large Scale Structure (GLASS)

mention the reason behind the choice of 30 shells.

3.2.2 The electron angular power spectrum

3.2.3 The cosmological covariance of the dispersion measure signal

3.3 The systematic uncertainties

It is important here to mention how this is modelled on the level of the simulation and the level of covariance.

3.3.1 Redshift uncertainty

3.3.2 Selection effects

3.3.3 Unlocalised hosts

3.3.4 Milkyway contribution

3.3.5 Host galaxy contribution

3.4 Simulation-based inference

3.4.1 Approximate Baysian inference

3.4.2 Neural Posterior Estimator

3.4.3 The Sequentual Neural Posterio Estimator

The truncated sequential neural posterior estimator

3.4.4 Flow matching posterior estimator

3.4.5 Data Compression

I simulate the electron density shells and place FRBs in the same redshift and angular position as in the mock catalog. This will reduce the computational cost and the need to calculate the covariance matrix for each realisation within the SBI pipeline. (Try to do this excercise without choosing the sdame redshifts for a lower number of sources such that it lowers down the computational costs)

Numerical derivative stability

The Sensitivity of compressed statistics to model parameters

Add a plot about derivative stability. Add a plot for the sensitivity.

CHAPTER 4

Results

CHAPTER 5

Discussion & Conclusion

APPENDIX A

Useful information

- Fourier transform conventions. - Virial theorem. - spherical collapse model and the critical density. - The Dirac delta function.

Bibliography

Adame, A. et al. (2025), *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics **2025** 021.

Aghanim, N. et al. (2020), *Planck 2018 results-VI. Cosmological parameters*, Astronomy & Astrophysics **641** A6.

Alonso, D. (2021), *Linear anisotropies in dispersion-measure-based cosmological observables*, Physical Review D **103**.

Asgari, M., A. J. Mead and C. Heymans (2023), *The halo model for cosmology: a pedagogical review*, arXiv preprint arXiv:2303.08752.

Bahcall, J. N. and L. Spitzer (1982), *The space telescope*, Scientific American **247** 40.

Baldauf, T. (2020), "Effective Field Theory of Large-Scale Structure", *Effective Field Theory in Particle Physics and Cosmology: Lecture Notes of the Les Houches Summer School: Volume 108, July 2017*, Oxford University Press, ISBN: 9780198855743, eprint: https://academic.oup.com/book/0/chapter/372565490/chapter-pdf/45818469/oso-9780198855743-chapter-7.pdf, URL: https://doi.org/10.1093/oso/9780198855743.003.0007.

Bartelmann, M. and P. Schneider (2001), *Weak gravitational lensing*, Physics Reports **340** 291.

Bennett, C. L. et al. (2003), *First-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: Foreground emission*, The Astrophysical Journal Supplement Series **148** 97.

Boggess, N. W. et al. (1992), *The COBE mission - its design and performance two years after launch*, The Astrophysical Journal **397** 420.

Bondi, H. and I. W. Roxburgh (2010), *Cosmology*, Courier Corporation.

Carroll, S. M. (2024), *Spacetime and geometry: An introduction to general relativity*, Cambridge University Press.

Crain, R. A. and F. van de Voort (2023), *Hydrodynamical simulations of the galaxy population: Enduring successes and outstanding challenges*, Annual Review of Astronomy and Astrophysics **61** 473.

Daalen, M. P. van, J. Schaye, C. Booth and C. Dalla Vecchia (2011), *The effects of galaxy formation on the matter power spectrum: a challenge for precision cosmology*, Monthly Notices of the Royal Astronomical Society **415** 3649.

Davis, M., J. Huchra, D. W. Latham and J. Tonry (1982), *A survey of Galaxy redshifts. II - the large scale space distribution*, The Astrophysical Journal **253** 423.

Davis, M. and P. J. Peebles (1983), *A survey of Galaxy redshifts. V - the two-point position and velocity correlations*, The Astrophysical Journal **267** 465.

Desjacques, V., D. Jeong and F. Schmidt (2018), *Large-scale galaxy bias*, Physics reports **733** 1.

Di Valentino, E. et al. (2025), *The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics*, Physics of the Dark Universe **49** 101965.

Dodelson, S. and F. Schmidt (2026), *Modern cosmology*, Academic Press.

Draine, B. T. (2011), Physics of the interstellar and Intergalactic Medium.

Eifler, T. et al. (2015), *Accounting for baryonic effects in cosmic shear tomography: Determining a minimal set of nuisance parameters using PCA*, Monthly Notices of the Royal Astronomical Society **454** 2451.

Hu, W. and M. Tegmark (1999), *Weak lensing: Prospects for measuring cosmological parameters*, The Astrophysical Journal **514**.

Hubble, E. (1929), *A relation between distance and radial velocity among extra-galactic nebulae*, Proceedings of the National Academy of Sciences **15** 168.

Jain, B. and A. Taylor (2003), *Cross-correlation tomography: Measuring dark energy evolution with weak lensing*, Physical Review Letters **91**.

Kaiser, N. and G. Squires (1993), *Mapping the dark matter with weak gravitational lensing*, The Astrophysical Journal **404** 441.

Konar, K., R. Reischke, S. Hagstotz, A. Nicola and H. Hildebrandt (2025), *Constraining the dispersion measure redshift relation with simulation-based inference*, The Open Journal of Astrophysics **8**.

Konietzka, R. M. et al. (2025), *Ray-tracing Fast Radio Bursts Through IllustrisTNG: Cosmological Dispersion Measures from Redshift 0 to 5.5*, arXiv: `2507.07090 [astro-ph.CO]`.

Kovač, M. et al. (2025), *Baryonification II: Constraining feedback with X-ray and kinematic Sunyaev-Zel'dovich observations*, Journal of Cosmology and Astroparticle Physics **2025** 046.

Krause, E. et al. (2021), *Dark energy survey year 3 results: Multi-probe modeling strategy and validation*, arXiv preprint arXiv:2105.13548.

Lesgourgues, J. (2011), *The cosmic linear anisotropy solving system (CLASS) I: Overview*, arXiv preprint arXiv:1104.2932.

Lorimer, D. R., M. Bailes, M. A. McLaughlin, D. J. Narkevic and F. Crawford (2007), *A bright millisecond radio burst of extragalactic origin*, Science **318** 777.

Luke Hoffmann, J. et al. (2024), *Modelling DSA, fast, and craft surveys in a z-dm analysis and constraining a minimum FRB Energy*, Publications of the Astronomical Society of Australia **42**.

Maoli, R. et al. (2001), *Cosmic Shear Analysis in 50 uncorrelated VLT Fields. implications for ω_0, σ_8*, Astronomy and Astrophysics **368** 766.

Maraio, A., A. Hall and A. Taylor (2025), *Mitigating baryon feedback bias in cosmic shear through a theoretical error covariance in the matter power spectrum*, Monthly Notices of the Royal Astronomical Society **537** 1749.

Mead, A. J., S. Brieden, T. Tröster and C. Heymans (2021), *hmcode-2020: improved modelling of non-linear cosmological power spectra with baryonic feedback*, Monthly Notices of the Royal Astronomical Society **502** 1401, ISSN: 0035-8711, eprint: `https://academic.oup.com/mnras/article-pdf/502/1/1401/42438464/stab082.pdf`, URL: `https://doi.org/10.1093/mnras/stab082`.

Moreira, M. G. et al. (2021), *Mitigating baryonic effects with a theoretical error covariance*, Monthly Notices of the Royal Astronomical Society **507** 5592.

Noether, E. (1983), *Invariante Variationsprobleme*, Springer Collected Works in Mathematics 231.

Petroff, E., J. W. Hessels and D. R. Lorimer (2019), *Fast radio bursts*, The Astronomy and Astrophysics Review **27**.

Piccirilli, G., M. Zennaro, C. García-García and D. Alonso (2025), *Robust cosmic shear with small-scale nulling*, Journal of Cosmology and Astroparticle Physics **2025** 017.

Pillepich, A. et al. (2018), *Simulating galaxy formation with the IllustrisTNG model*, Monthly Notices of the Royal Astronomical Society **473** 4077.

Planck, M. (1900), *Zur Theorie des Gesetzes der Energieverteilung im Normalspektrum*, Verhandlungen der Deutschen Physikalischen Gesellschaft **2** 237.

Pope, A. (2004), *Cosmological parameters from Eigenmode Analysis of Sloan Digital Sky Survey Galaxy Redshifts*, AIP Conference Proceedings **743** 120.

Reischke, R. and S. Hagstotz (2025), *A first measurement of baryonic feedback with Fast Radio Bursts*, arXiv: 2507.17742 [astro-ph.CO].

Reischke, R., D. Neumann, K. A. Bertmann, S. Hagstotz and H. Hildebrandt (2023), *Calibrating baryonic feedback with weak lensing and fast radio bursts*, arXiv preprint arXiv:2309.09766.

Riess, A. G., W. Yuan et al. (2022), *A comprehensive measurement of the local value of the Hubble constant with 1 km s- 1 Mpc- 1 uncertainty from the Hubble Space Telescope and the SH0ES team*, The Astrophysical journal letters **934** L7.

Riess, A. G., A. V. Filippenko et al. (1998), *Observational evidence from supernovae for an accelerating universe and a cosmological constant*, The Astronomical Journal **116** 1009.

Rudd, D. H., A. R. Zentner and A. V. Kravtsov (2008), *Effects of baryons and dissipation on the matter power spectrum*, The Astrophysical Journal **672** 19.

Schneider, A., M. Kovač et al. (2025), *Baryonification: An alternative to hydrodynamical simulations for cosmological studies*, Journal of Cosmology and Astroparticle Physics **2025** 043.

Schneider, A. and R. Teyssier (2015), *A new method to quantify the effects of baryons on the matter power spectrum*, Journal of Cosmology and Astroparticle Physics **2015** 049.

Schneider, P., T. Eifler and E. Krause (2010), *COSEBIs: Extracting the full E-/B-mode information from cosmic shear correlation functions*, Astronomy and Astrophysics **520**.

Schneider, P., J. Ehlers and E. E. Falco (1992), *Gravitational lenses*, Springer International Publishing.

Schneider, P., L. Van Waerbeke, B. Jain and G. Kruse (1998), *A new measure for Cosmic Shear*, Monthly Notices of the Royal Astronomical Society **296** 873.

Seitz, C. and P. Schneider (1996), *Steps towards nonlinear cluster inversion through gravitational distortions: III. Including a redshift distribution of the sources*, arXiv preprint astro-ph/9601079.

Sharma, K. et al. (2026), *Probing baryonic feedback and cosmology with the 3 × 2-point statistic of fast radio bursts and galaxies*, The Astrophysical Journal **998** 109.

Sitter, W. de (1917), *On Einstein's theory of gravitation and its astronomical consequences. third paper.*, Monthly Notices of the Royal Astronomical Society **78** 3.

Smoot, G. F. (1999), *COBE observations and results*, Conference on 3K cosmology 1.

Somerville, R. S. and R. Davé (2015), *Physical models of galaxy formation in a cosmological framework*, Annual Review of Astronomy and Astrophysics **53** 51.

Takahashi, R. (2024), *Impact of lensing bias on the cosmological dispersion measure*, Journal of Cosmology and Astroparticle Physics **2024** 057.

Tinker, J. L. et al. (2010), *The large-scale bias of dark matter halos: Numerical calibration and model tests*, The Astrophysical Journal **724** 878.

Torkamani, M. et al. (2026), *Baryonification III: An accurate analytical model for the dispersion measure probability density function of fast radio bursts*, arXiv preprint arXiv:2601.18784.

Totsuji, H. and T. Kihara (1969), *The correlation function for the distribution of galaxies*, Publications of the Astronomical Society of Japan **21** 221.

Turner, M. S. (2001), *Dark matter and dark energy in the universe*, Particle Physics and the Universe 210.

Verde, L., N. Schöneberg and H. Gil-Marín (2024), *A tale of many H 0*, Annual Review of Astronomy and Astrophysics **62** 287.

Wechsler, R. H. and J. L. Tinker (2018), *The connection between galaxies and their dark matter halos*, Annual Review of Astronomy and Astrophysics **56** 435.

List of Figures

2.1 A schematic representation of spacetime that follows Weyl's postulate. The green lines labelled with letters are world lines, the path of freely falling objects in the spacetime (or geodesics). The redlines are spacelike hypersurfaces labelled with the same cosmic time t_i. These hypersuraces are homogeneous and isotropic. 9
2.2 A schematic representation of density fractions of the different components in the ΛCDM model. The density fractions are measured by the Planck collaboration (Aghanim et al., 2020) using measurements of the CMB anisotropies. 16
2.3 A 2D plot from the IllustrisTNG simulation (Pillepich et al., 2018), adopted from Crain and Voort (2023). It is a snapshot of hydrodynamical simulation showing how a 'danser than average' region of dark matter hosts a galaxy (stars make up the 'skeleton' of a galaxy). 19

List of Tables

Acknowledgements

I am strongly indebted to my supervisors Dr. Andrina Nicola and Dr. Robert Reischke for their immense support throughout the duration my master's studies. Their support and advice were pivotal in the process of conducting this work. I would also like to thank the member of the 'Combining Cosmological Probes' group at the Argelander Institute for Astronomy, especially, Subarna, Mokka, Micheal, Reihana, and Reiko.

I would like to thank Prof. Dr. Jennifer Schöber for stepping in to be my supervisor for the second part of my thesis.

Furthermore, I would like to thank my friends and colleagues at the University of Bonn who made the master's studies journey more fun and less stressful. I would like to thank my aunt, Sana, who kept my company and offered immense support when adjusting to the life in Germany and Bonn.

I thank my girlfriend, Fée, who was my best friend throughout the journey of my master's thesis. Her words of encourgments were crucial in the moments of doubt.

I would further like to thank the memebers of the physical cosmology department at the Max Planck Institute for Astrophysics where part of this work was conducted, in particular, Fabian Schmidt, Šafak Celik, Lena Scheuchel, Sveva Castillo, Florian List, and Adr

In addition, I would like to thank Dr. Supranta Boruah and Dr. Gary Bernstein for offering me the opportunity to work with them on the topic of field-level inference and empirical modelling of galaxy bias. Though not complete, working with them on this project improved the way I approach and think about problems in cosmology.